COIL RAIL - 4.0L (Continued)

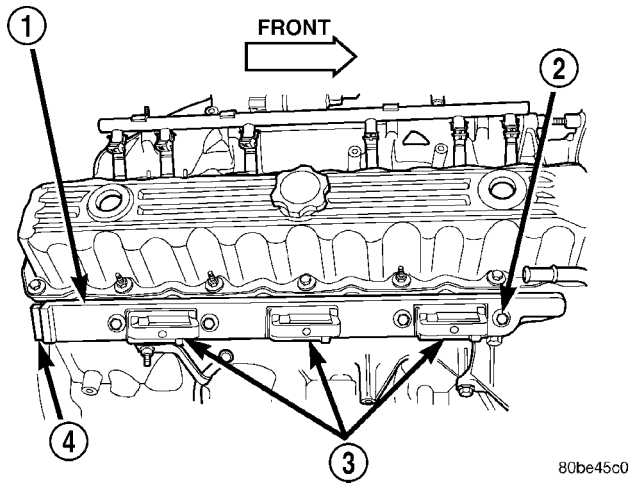

Fig. 26 Ignition Coil Rail Location—4.0L 6–Cylinder Engine

1 - COIL RAIL
2 - COIL MOUNTING BOLTS (4)
3 - COIL
4 - COIL ELECTRICAL CONNECTION

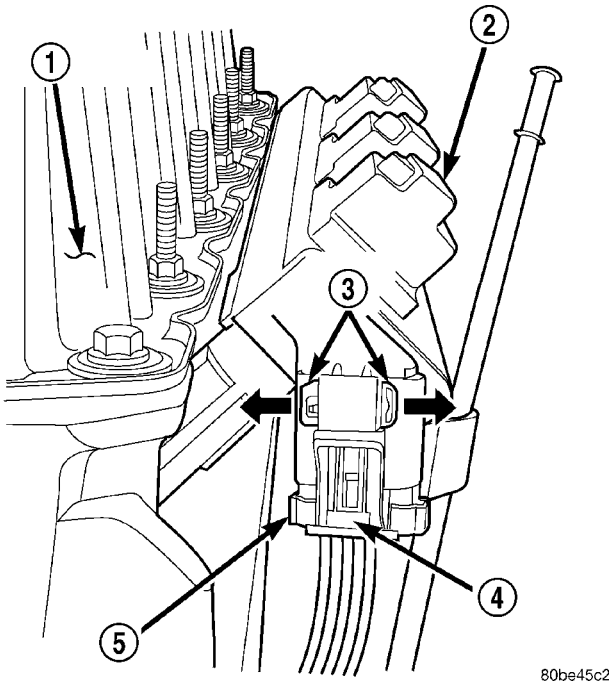

Fig. 27 Ignition Coil Electrical Connector—4.0L 6–Cylinder Engine

1 - REAR OF VALVE COVER
2 - COIL RAIL
3 - SLIDE TAB
4 - RELEASE LOCK
5 - COIL CONNECTOR

INSTALLATION - 4.0L

(1) Connect engine harness connector to coil by snapping into position. Move slide tab towards engine (Fig. 27) for a positive lock.

(2) Position ignition coil rubber boots to all spark plugs. Push down on coil assembly until bolt bases have contacted cylinder head

(3) Install 4 coil mounting bolts. Loosely tighten 4 bolts just enough to allow bolt bases to contact cylinder head. Do a final tightening of each bolt in steps down to 29 N·m (250 in. lbs.) torque. Do not apply full torque to any bolt first.

(4) Connect negative battery cable to battery.

IGNITION COIL - 2.4L

DESCRIPTION - 2.4L

The coil assembly consists of 2 different coils molded together. The assembly is mounted to the top of the engine (Fig. 28).

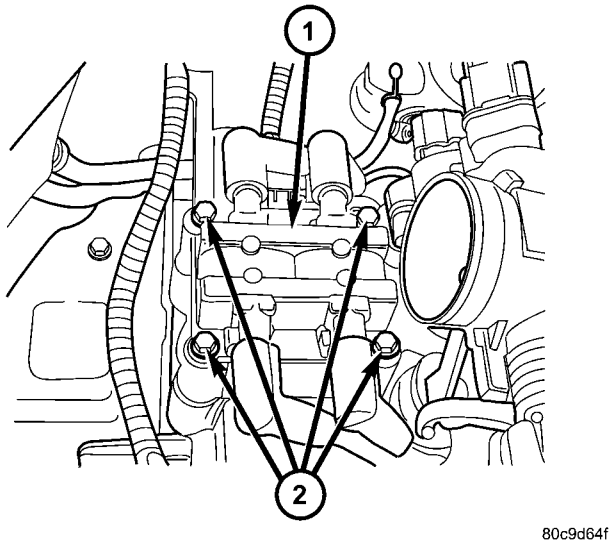

Fig. 28 IGNITION COIL - 2.4L

1 - IGNITION COIL
2 - MOUNTING BOLTS (4)

OPERATION - 2.4L

The coil fires two spark plugs simultaneously. One plug is under compression, the other plug fires on the exhaust stroke (lost spark). Coil number one fires cylinders 1 and 4, and coil number two fires cylinders 2 and 3.

The Auto Shutdown (ASD) relay provides battery voltage to the ignition coil. The PCM provides a ground contact (circuit) for energizing the coil(s). The PCM will de-energize the ASD relay if it does not receive the crankshaft position sensor and camshaft position sensor inputs.

Base ignition timing is not adjustable. By controlling the coil ground circuit, the PCM is able to set the base timing and adjust the ignition timing advance. This is done to meet changing engine operating conditions.